

Model Testing Report

Experiment: output_pandas_py39

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1) Model Effectiveness (1/3)

Metric	Figure	Explanation
auc_roc	0.8299	Measures ranking power across all thresholds. 0.8299 indicates strong separation. Next: compare AUC-ROC across segments and keep threshold tuning separate from ranking validation.
auc_pr	0.6833	Measures precision-recall performance under class imbalance; the observed event rate is 5.76%. Next: use this together with alert capacity to decide whether the high-score queue is practically useful.
precision	0.9666	Shows the share of flagged cases that are true positives at the current threshold. Next: if this is below review expectations, raise the threshold or improve exclusion features.
recall	0.6035	Shows the share of true positives captured at the current threshold. Next: if missed cases are too high, lower the threshold or add stronger risk features.
f1	0.7431	Balances precision and recall in one score. Next: use it as a threshold-selection aid when both missed cases and false alerts matter.
ks_statistic	0.6422	Measures score separation between positive and negative classes. Next: low KS suggests the model needs better features or a different specification.

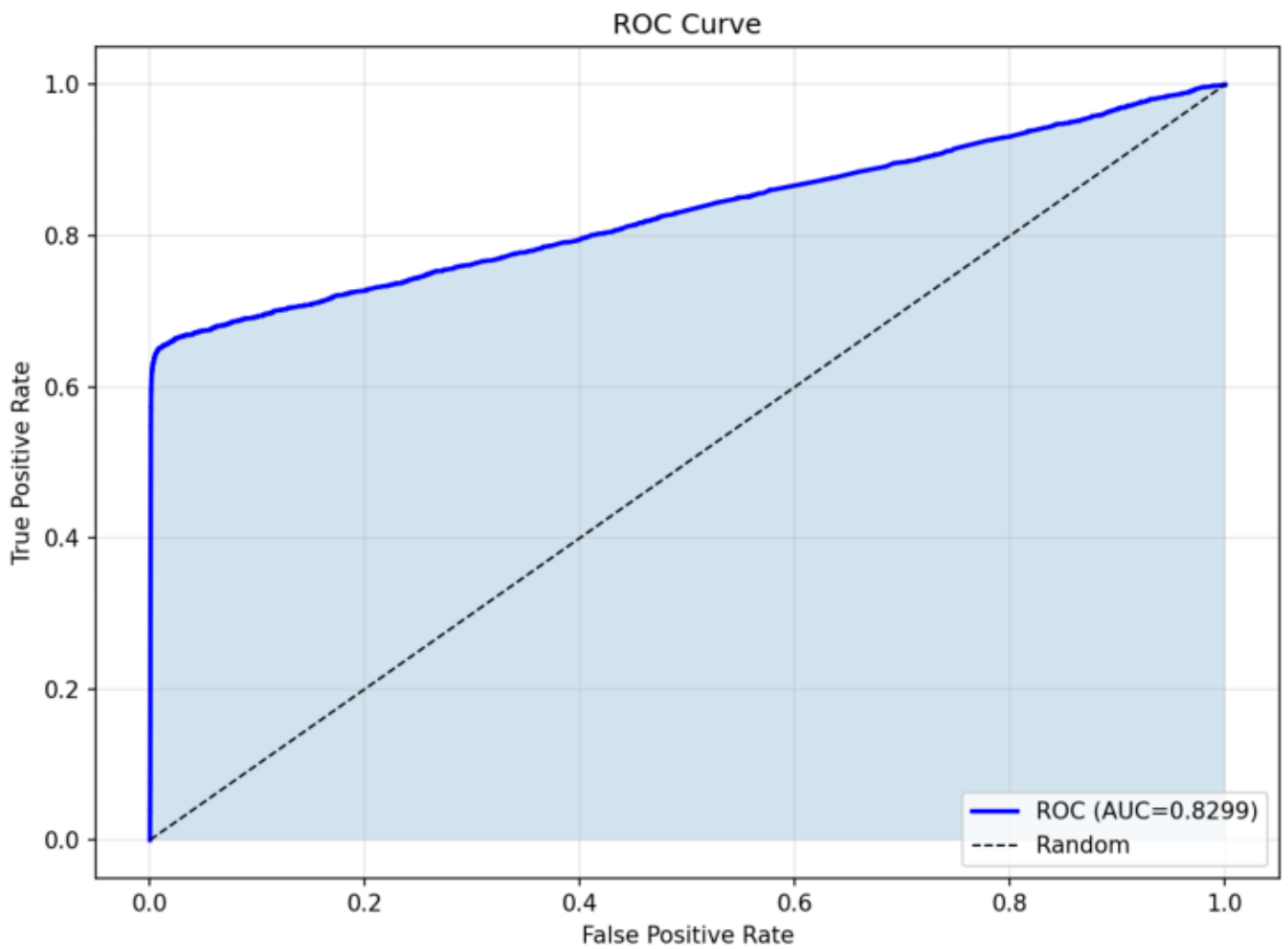
1) Model Effectiveness (2/3)

Metric	Figure	Explanation
ks_threshold	0.1200	This is the threshold that maximizes KS separation. Next: treat it as a candidate operating point, then compare it with business capacity and policy limits.
confusion_matrix[TN]	56473	True negatives correctly cleared. Next: preserve this base while tuning the alert threshold.
confusion_matrix[FP]	72	False positives create review cost without SAR conversion. Next: reduce this count with threshold tuning or sharper risk features.
confusion_matrix[FN]	1370	False negatives are missed SAR cases. Next: if this count is unacceptable, lower the threshold or add features that capture suspicious behavior earlier.
confusion_matrix[TP]	2085	True positives are correctly escalated cases. Next: assess whether this volume is sufficient relative to expected SAR capture goals.
precision_at_k[10]	1.0000	Shows how accurate the top-10 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[50]	0.9600	Shows how accurate the top-50 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[100]	0.9600	Shows how accurate the top-100 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.

1) Model Effectiveness (3/3)

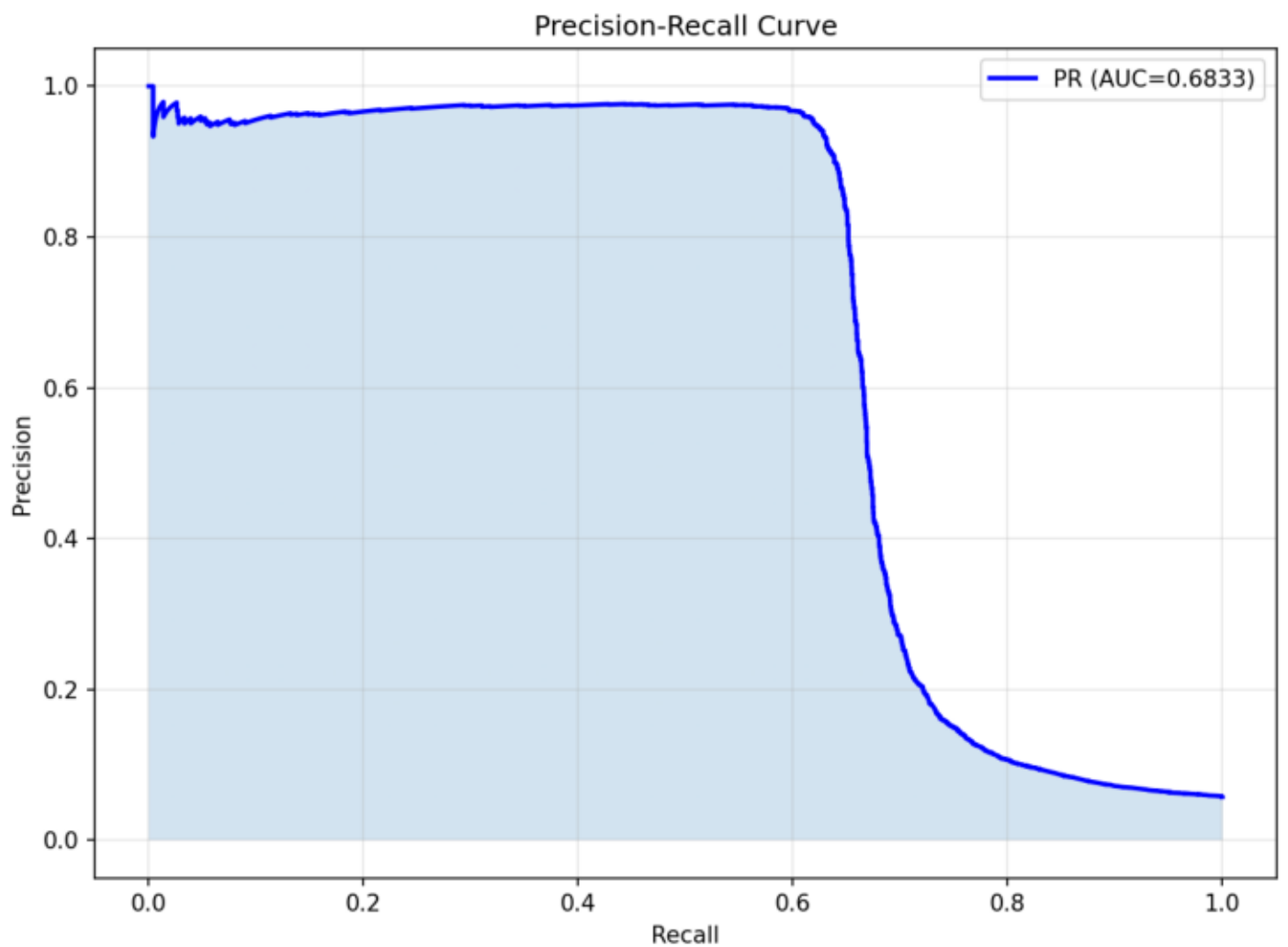
Metric	Figure	Explanation
precision_at_k[200]	0.9500	Shows how accurate the top-200 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
precision_at_k[500]	0.9620	Shows how accurate the top-500 highest-risk queue is. Next: align K with investigator capacity and confirm that top alerts stay high quality.
recall_at_k[10]	0.0029	Shows how much of the positive population is captured within the top-10 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[50]	0.0139	Shows how much of the positive population is captured within the top-50 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[100]	0.0278	Shows how much of the positive population is captured within the top-100 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[200]	0.0550	Shows how much of the positive population is captured within the top-200 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.
recall_at_k[500]	0.1392	Shows how much of the positive population is captured within the top-500 queue. Next: increase K if coverage is too low, or improve ranking quality if capacity cannot expand.

1) Model Effectiveness



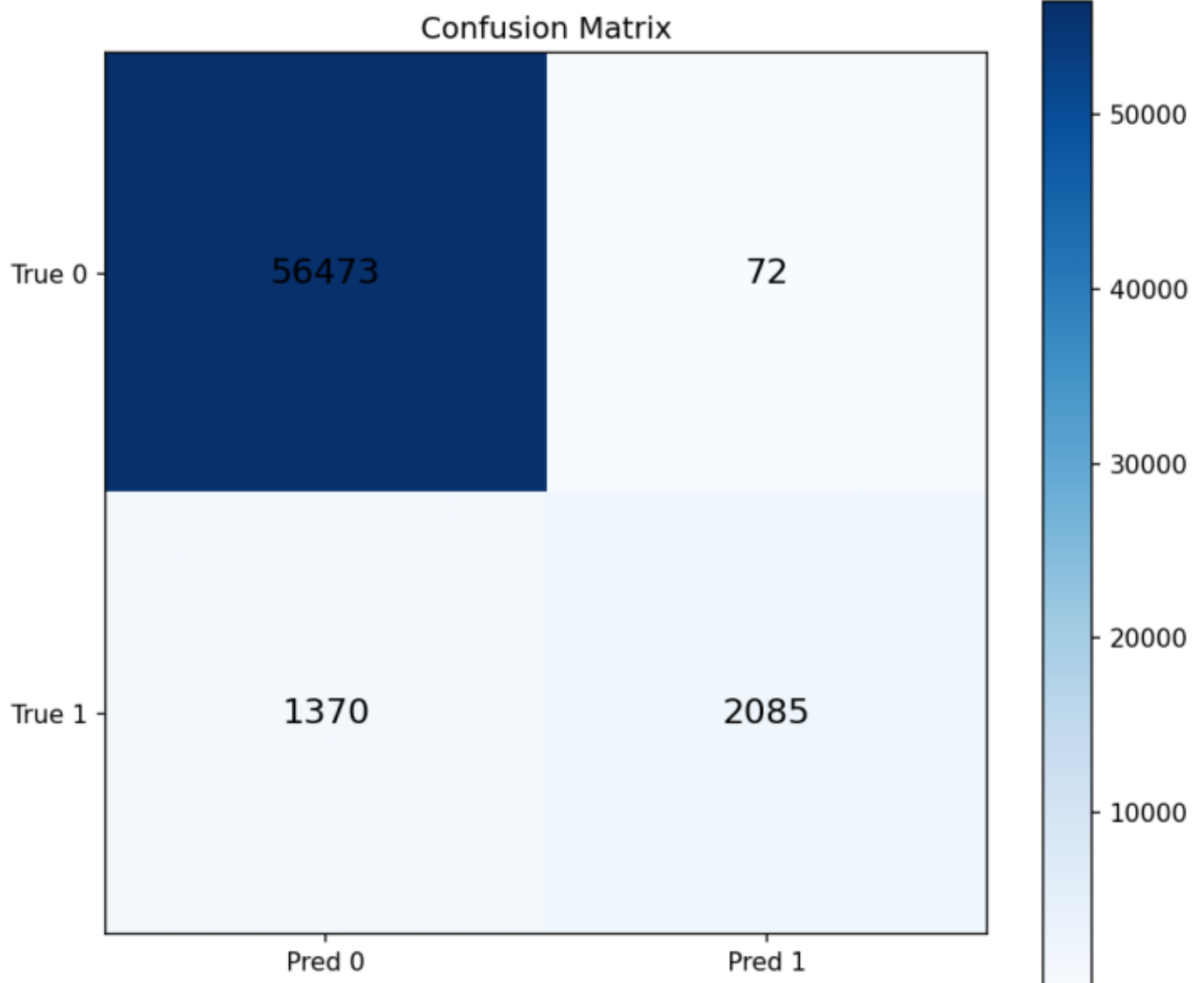
roc_curve: AUC-ROC=0.8299; curve above diagonal indicates strong separation.

1) Model Effectiveness



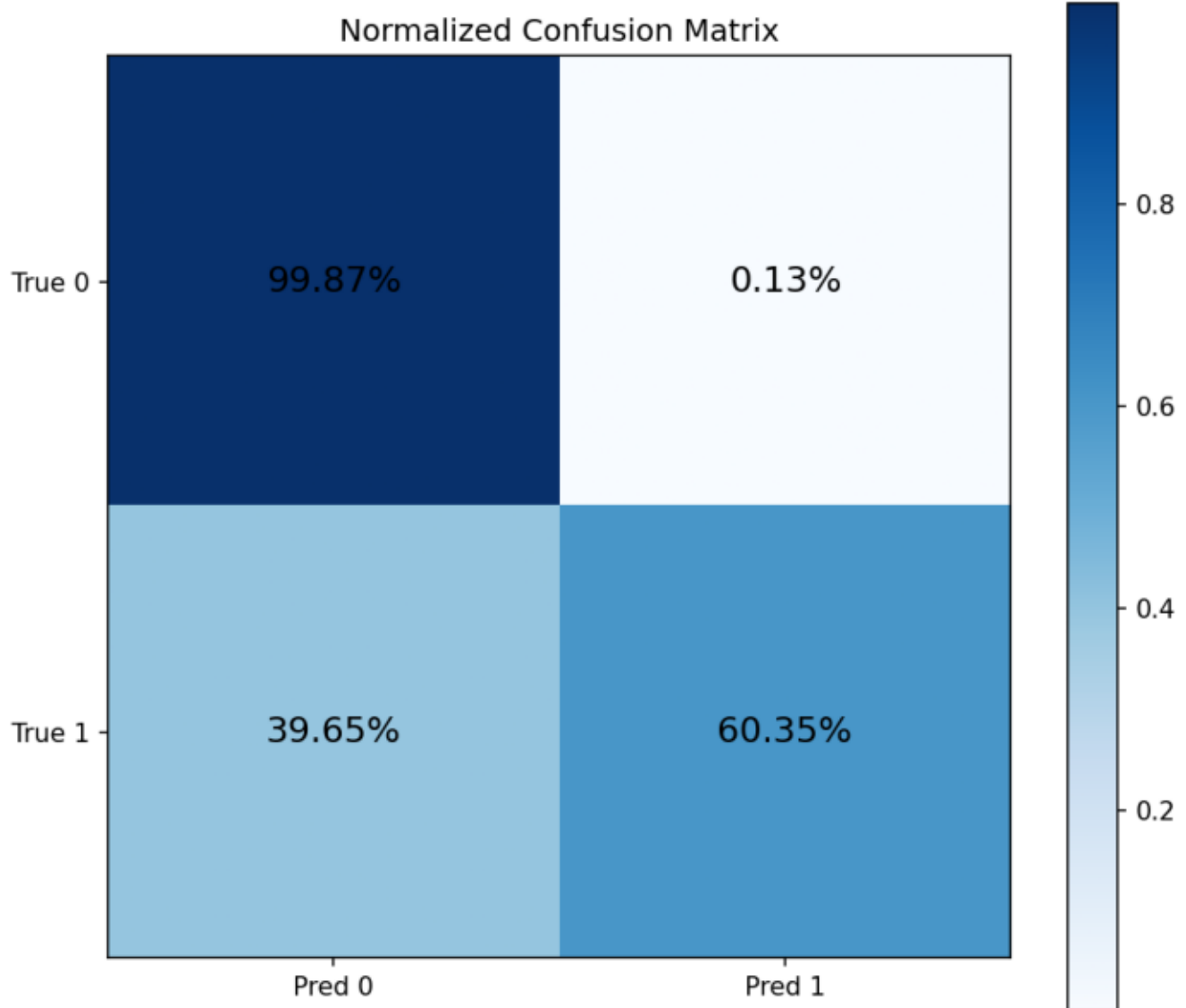
pr_curve: AUC-PR=0.6833 vs baseline 5.76%; higher curve implies better precision at recall.

1) Model Effectiveness



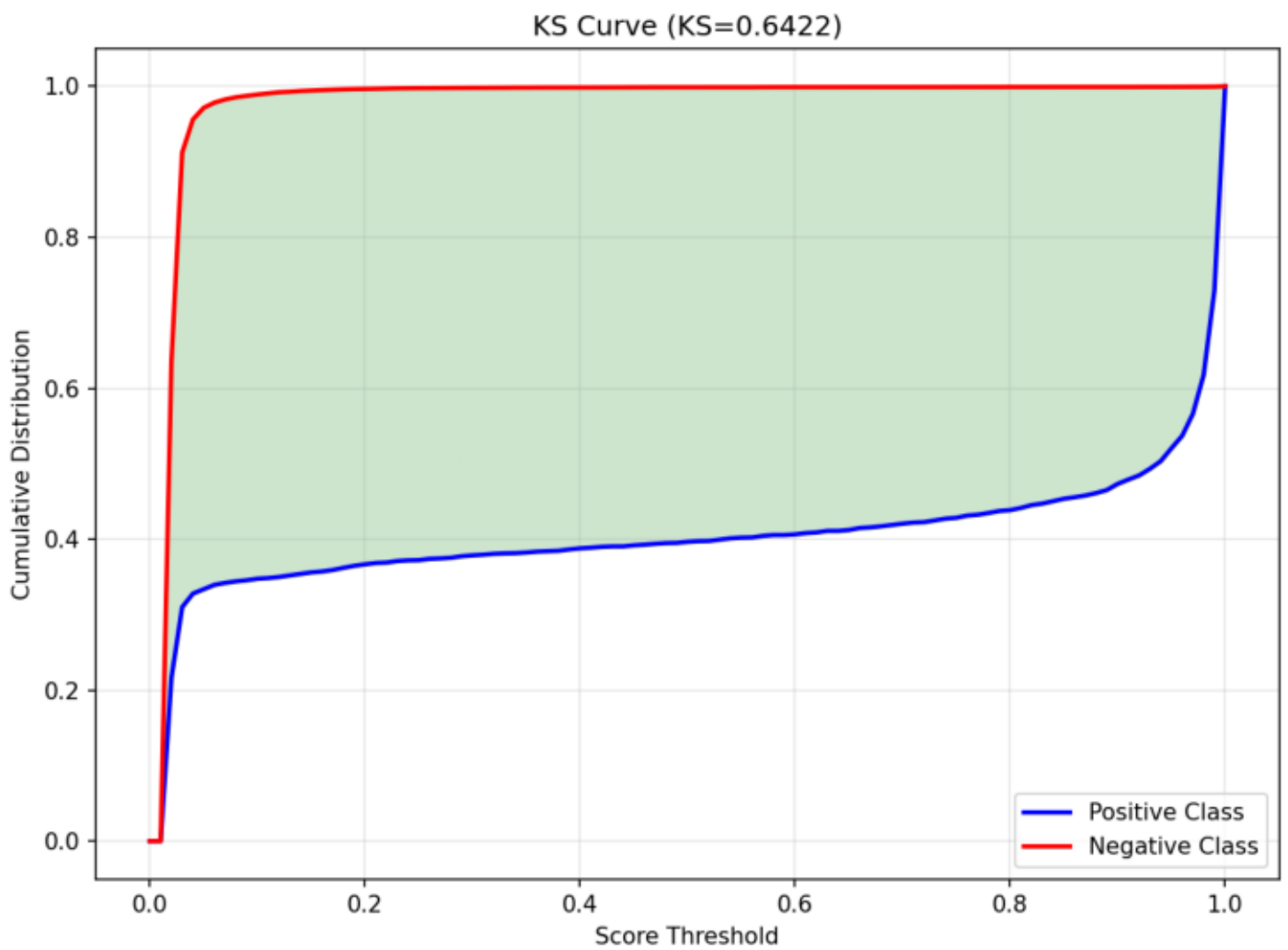
confusion_matrix: At threshold 0.50: TP=2085, FP=72, FN=1370, TN=56473.

1) Model Effectiveness



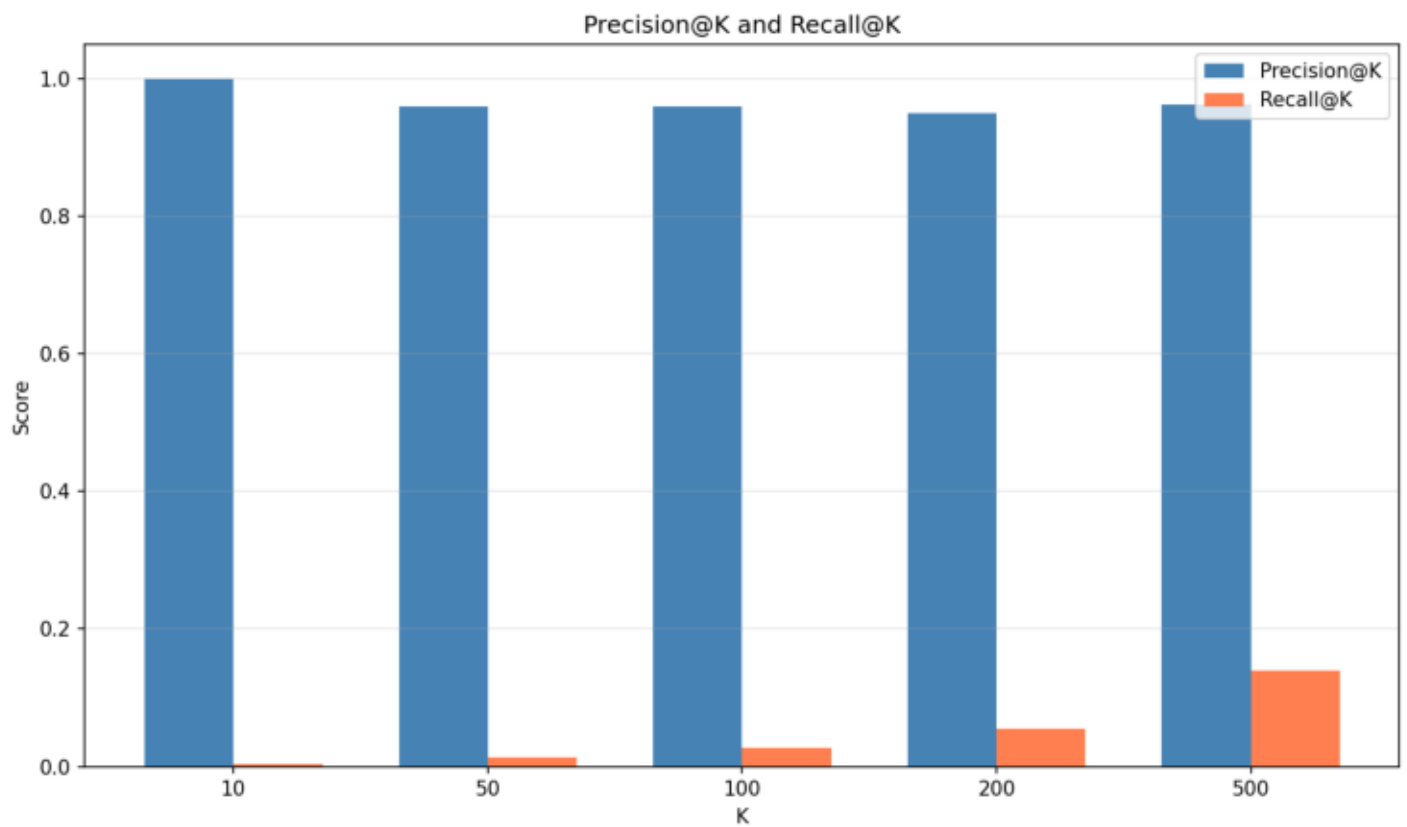
confusion_matrix_norm: Normalized rates: TPR=60.35%, TNR=99.87%.

1) Model Effectiveness



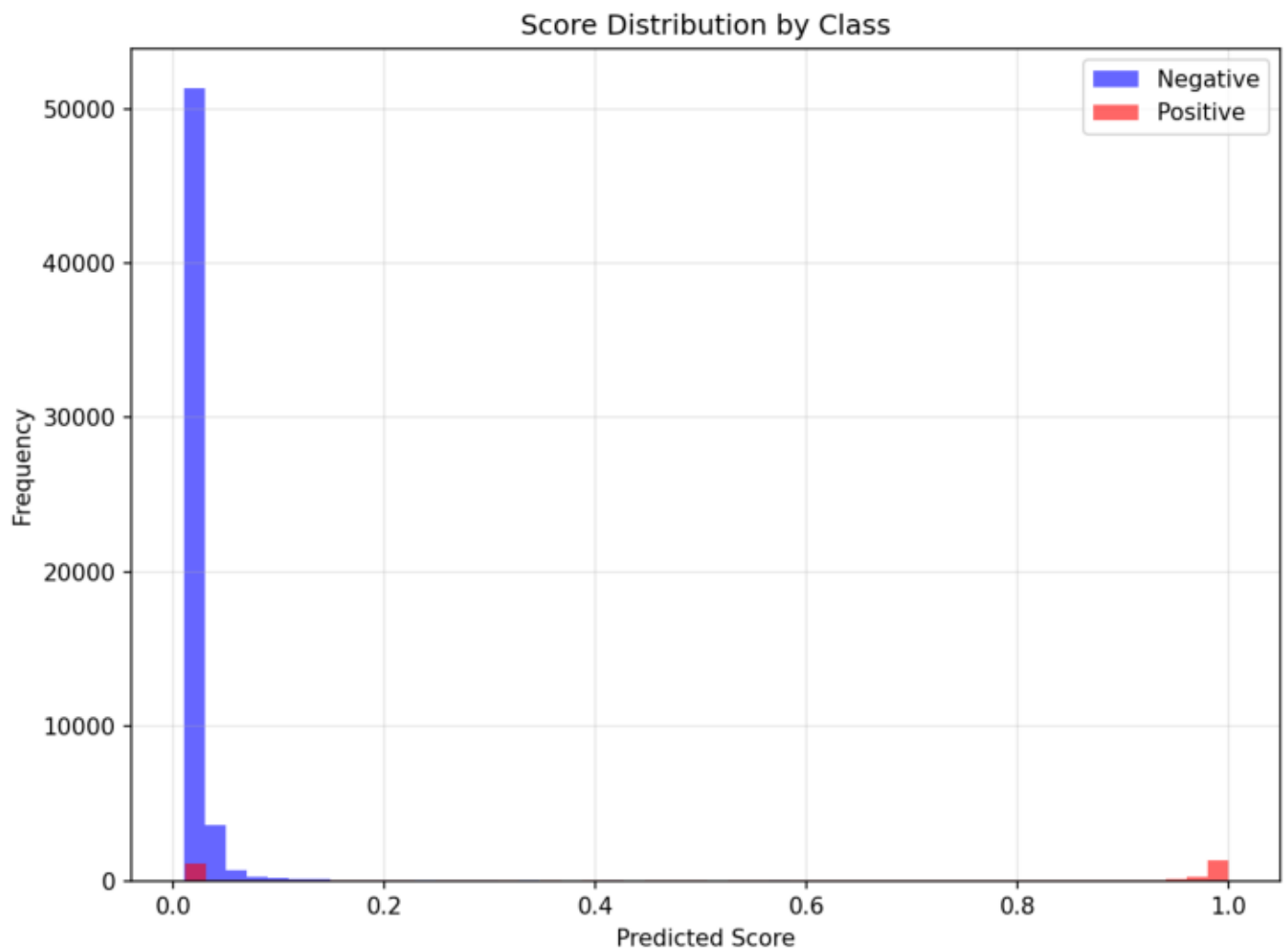
ks_curve: Maximum separation KS=0.6422 at threshold 0.12.

1) Model Effectiveness



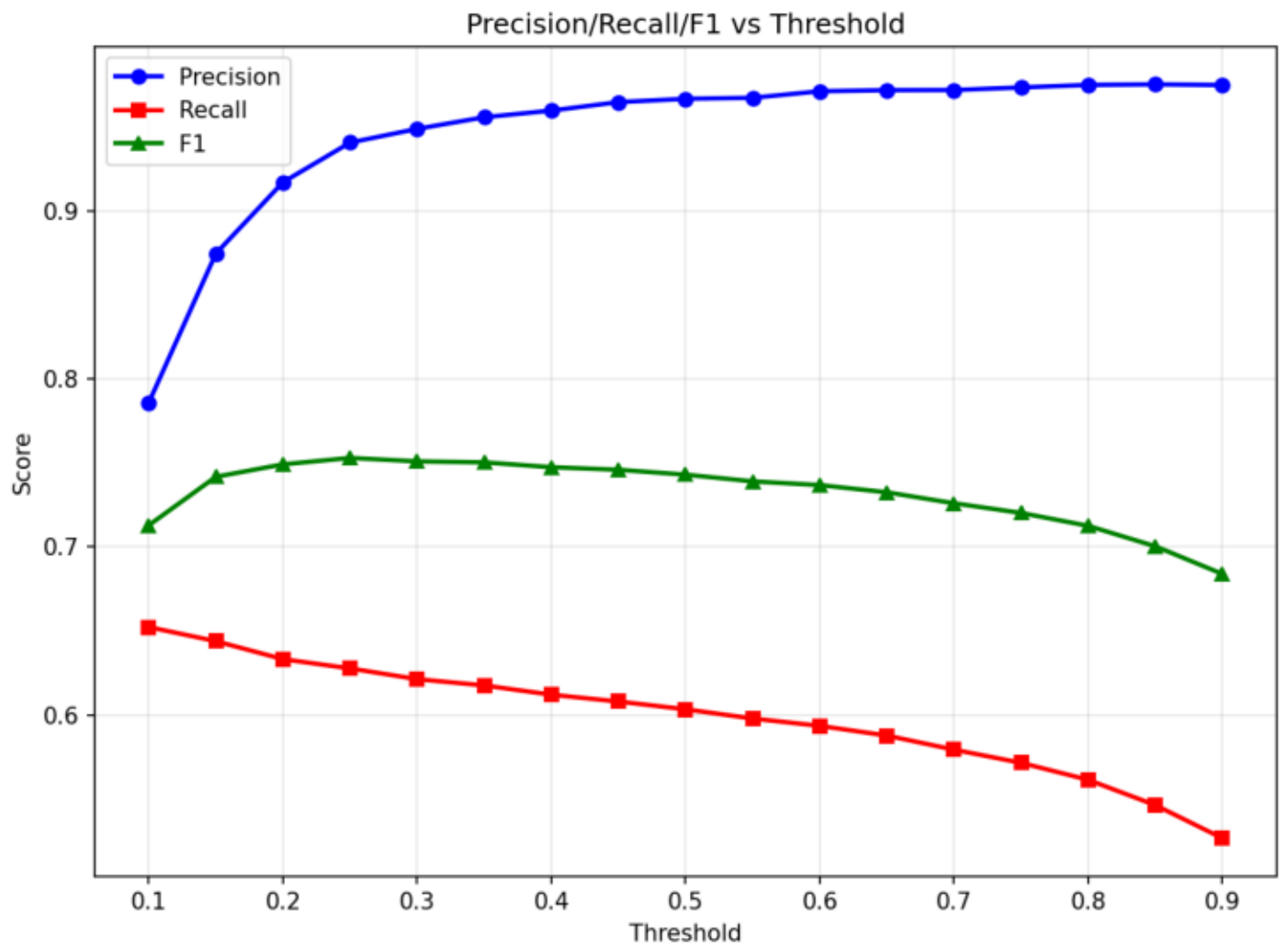
precision_recall_at_k: Bars show precision/recall as you expand the review queue.

1) Model Effectiveness



score_distribution: Mean score: positive=0.597, negative=0.024 (larger gap is better).

1) Model Effectiveness



threshold_analysis: Best F1 in tested grid is 0.7530 at threshold 0.25.

2) Model Efficiency (1/4)

Metric	Figure	Explanation
fpr	0.0013	Measures the share of negatives incorrectly flagged; 0.0013 is low. Next: if review demand is too high, raise the threshold or add features that better suppress false alerts.
tn	56473	Counts negatives correctly cleared without manual review. Next: preserve this clearance rate while checking that recall does not fall below policy expectations.
fp	72	Counts non-events that would still be sent to review. Next: use this with staffing capacity to decide whether threshold or feature refinement is required.
threshold	0.5000	This is the active operating cutoff used to flag alerts. Next: tune it jointly with recall, precision, and review capacity rather than in isolation.
fpr_at_thresholds[thresholds.t_0.05]	0.0285	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.05. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.10]	0.0109	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.10. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

2) Model Efficiency (2/4)

Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.15]	0.0056	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.15. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.20]	0.0035	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.20. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.25]	0.0024	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.25. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.30]	0.0021	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.30. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.35]	0.0018	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.35. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.40]	0.0016	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.40. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

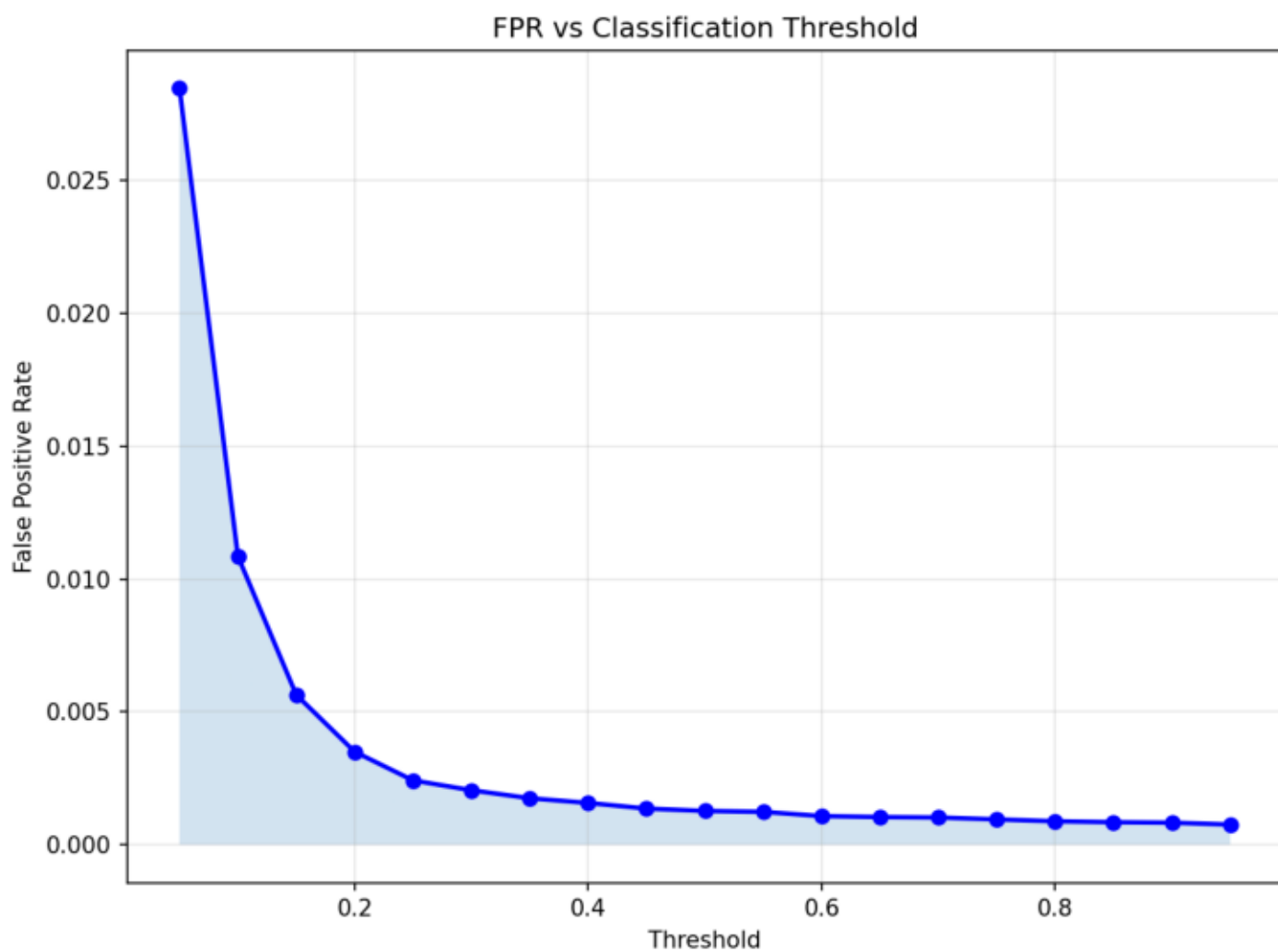
2) Model Efficiency (3/4)

Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.45]	0.0014	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.45. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.50]	0.0013	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.50. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.55]	0.0012	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.55. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.60]	0.0011	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.60. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.65]	0.0010	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.65. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.70]	0.0010	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.70. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

2) Model Efficiency (4/4)

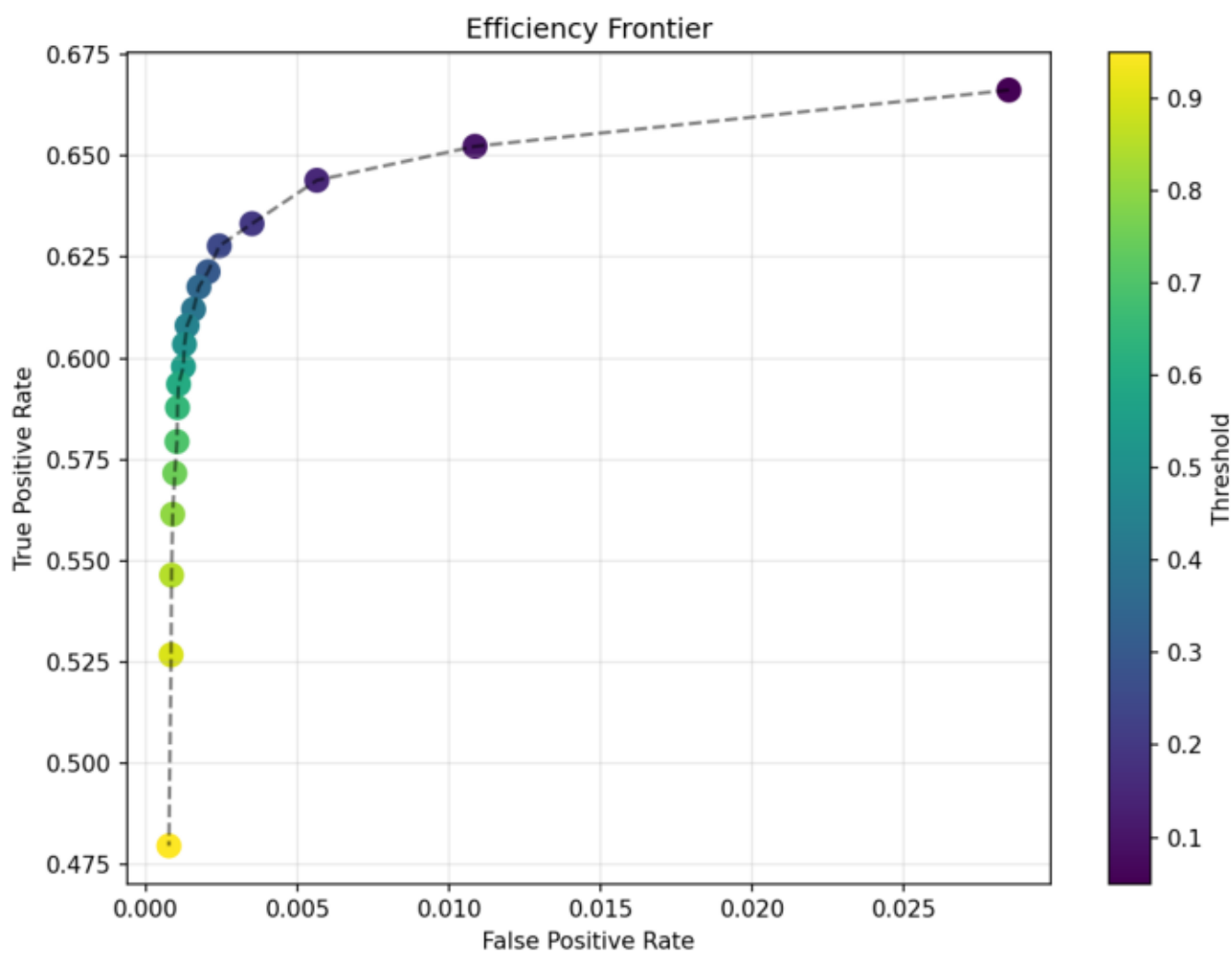
Metric	Figure	Explanation
fpr_at_thresholds[thresholds.t_0.75]	0.0010	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.75. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.80]	0.0009	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.80. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.85]	0.0008	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.85. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.90]	0.0008	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.90. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.
fpr_at_thresholds[thresholds.t_0.95]	0.0008	Shows the expected false-positive rate if the threshold were set to thresholds.t_0.95. Next: use these candidate cutoffs to choose a reviewable alert volume before production sign-off.

2) Model Efficiency



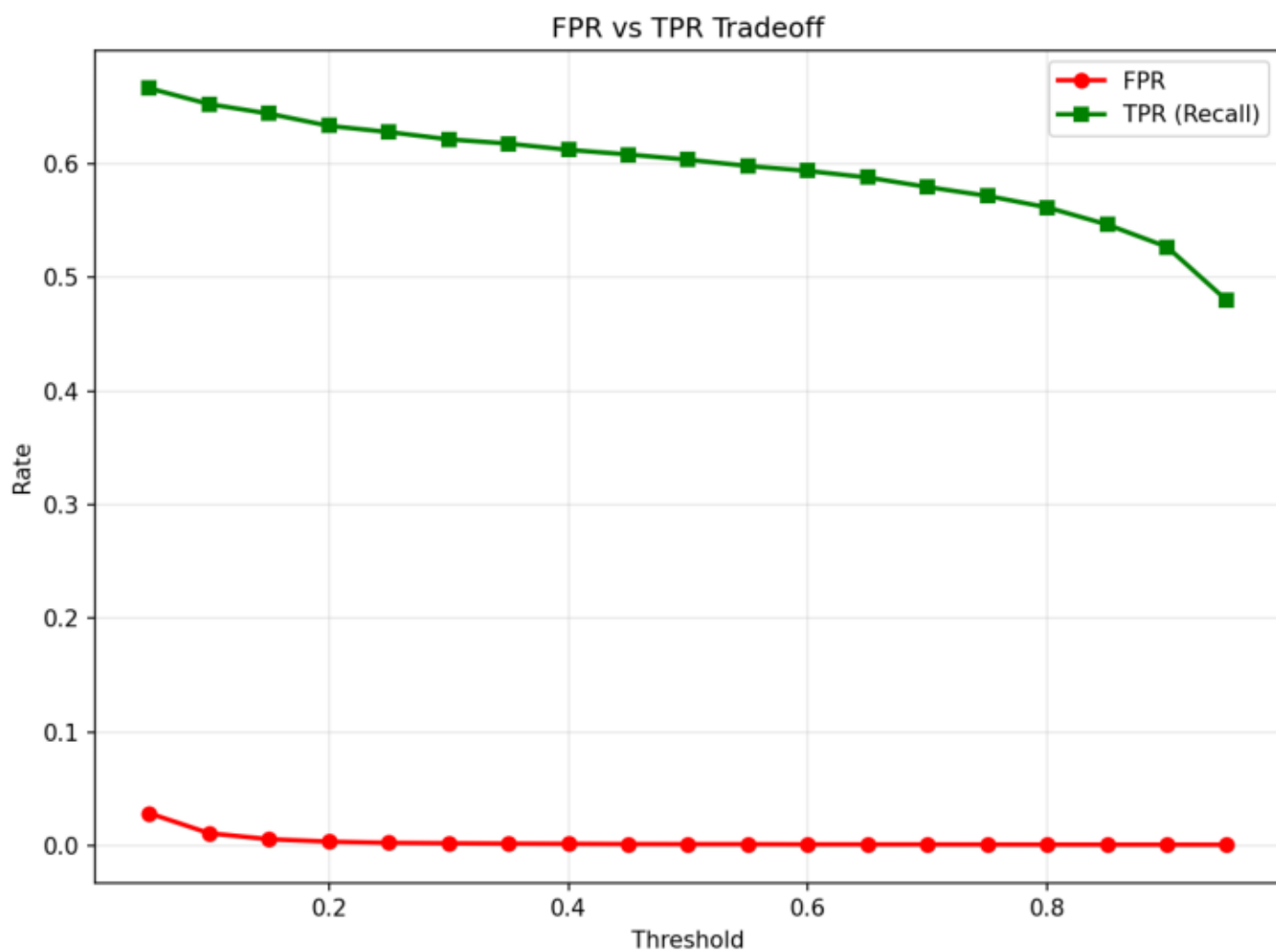
fpr_vs_threshold: FPR decreases as the threshold increases; use it to pick an operating point.

2) Model Efficiency



efficiency_frontier: Each point shows the FPR/TPR tradeoff; move toward the top-left for better efficiency.

2) Model Efficiency



fpr_tpr_tradeoff: FPR and TPR curves highlight how recall drops as you reduce false positives.

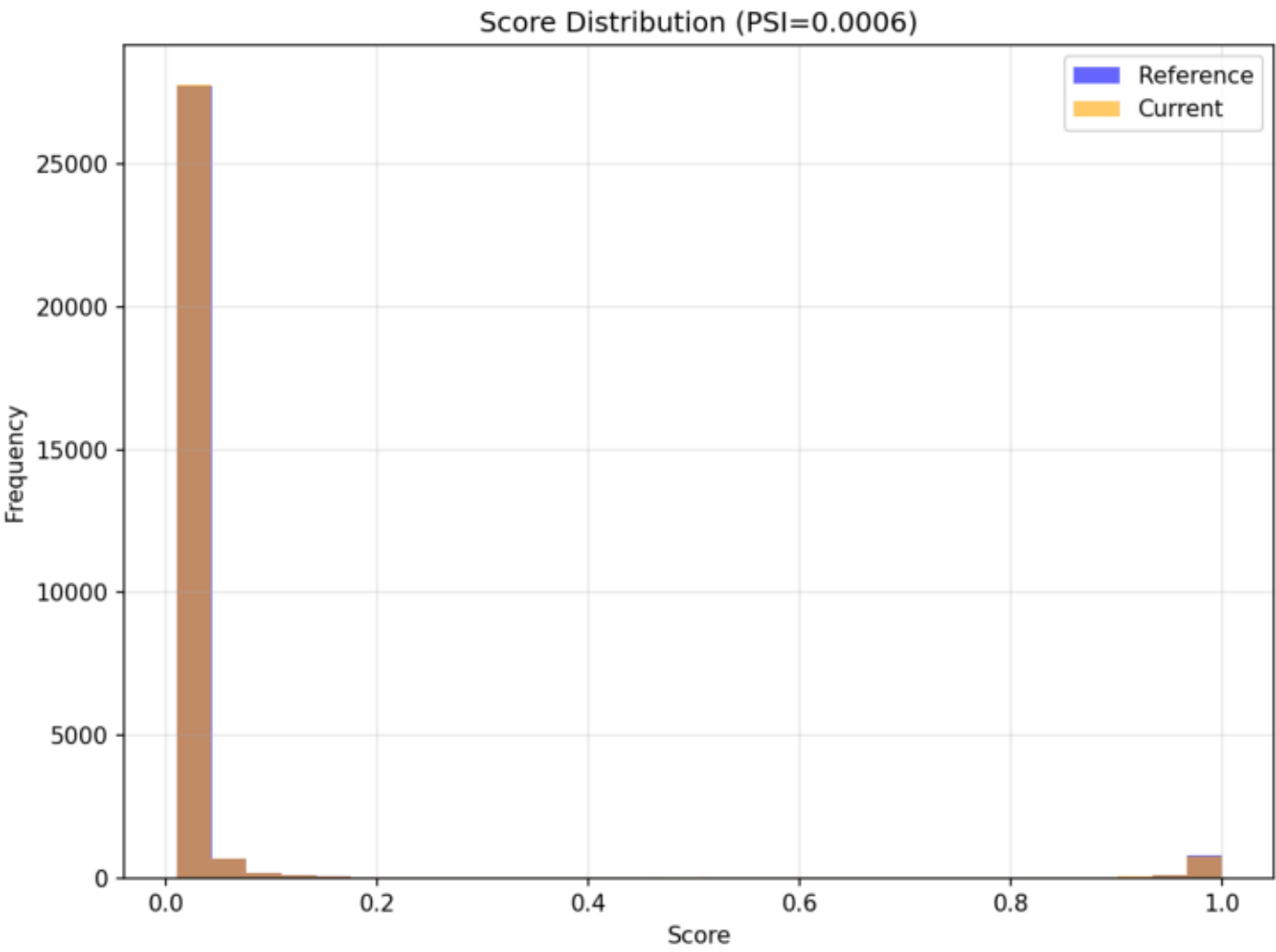
3) Model Stability (1/2)

Metric	Figure	Explanation
psi	0.0006	Measures score-distribution shift between reference and current populations; 0.0006 is negligible. Next: investigate segment mix, data drift, and recalibration needs when PSI is material.
cv_auc_roc_mean	0.8298	Average AUC-ROC across validation folds. Next: compare it with overall AUC-ROC; a large gap can indicate overfitting or unstable sampling.
cv_auc_roc_std	0.0107	Standard deviation of AUC-ROC across folds. Next: if variability is high, simplify the model or increase training stability before deployment.
cv_auc_pr_mean	0.6837	Average AUC-PR across validation folds. Next: confirm that precision-recall performance remains acceptable under resampling.
cv_auc_pr_std	0.0088	Standard deviation of AUC-PR across folds. Next: high dispersion suggests unstable alert quality and warrants deeper validation.
bootstrap_auc_roc_mean	0.8357	Bootstrap mean AUC-ROC estimates expected ranking performance under repeated sampling. Next: compare it with the point estimate to judge optimism in the current test run.
bootstrap_auc_roc_ci_lower	0.8238	Lower confidence bound for bootstrap AUC-ROC. Next: use it as a conservative performance floor in model-risk discussions.
bootstrap_auc_roc_ci_upper	0.8467	Upper confidence bound for bootstrap AUC-ROC. Next: review the interval width with the lower bound to assess uncertainty.

3) Model Stability (2/2)

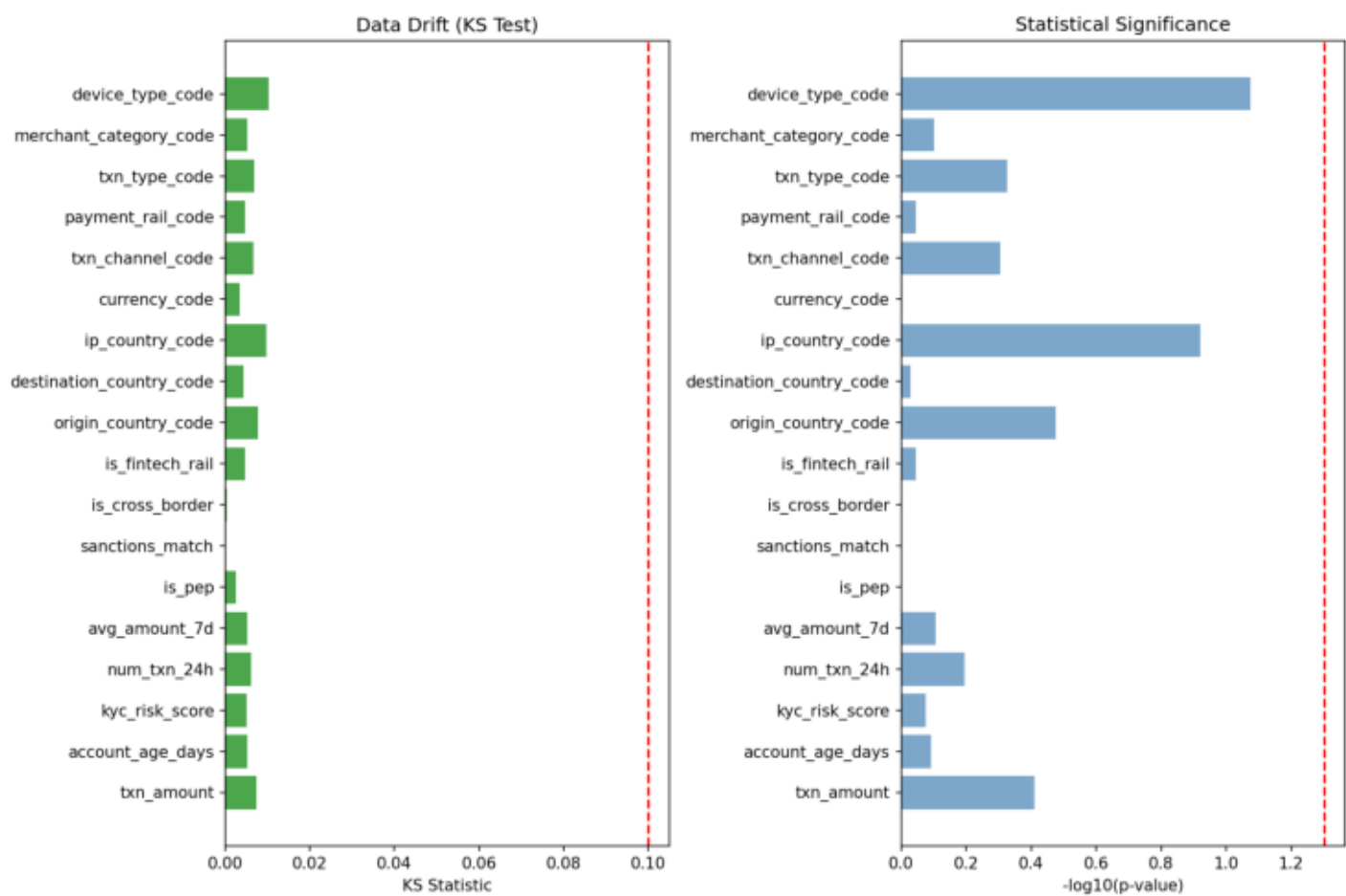
Metric	Figure	Explanation
concept_drift_detected	False	No material concept-drift signal was detected in the evaluated slices. Next: retain ongoing monitoring because stable historical results do not guarantee future stability.
concept_drift_score	0.0225	Quantifies the magnitude of performance change across ordered slices. Next: rising scores should trigger deeper review of segment behavior and model refresh timing.

3) Model Stability



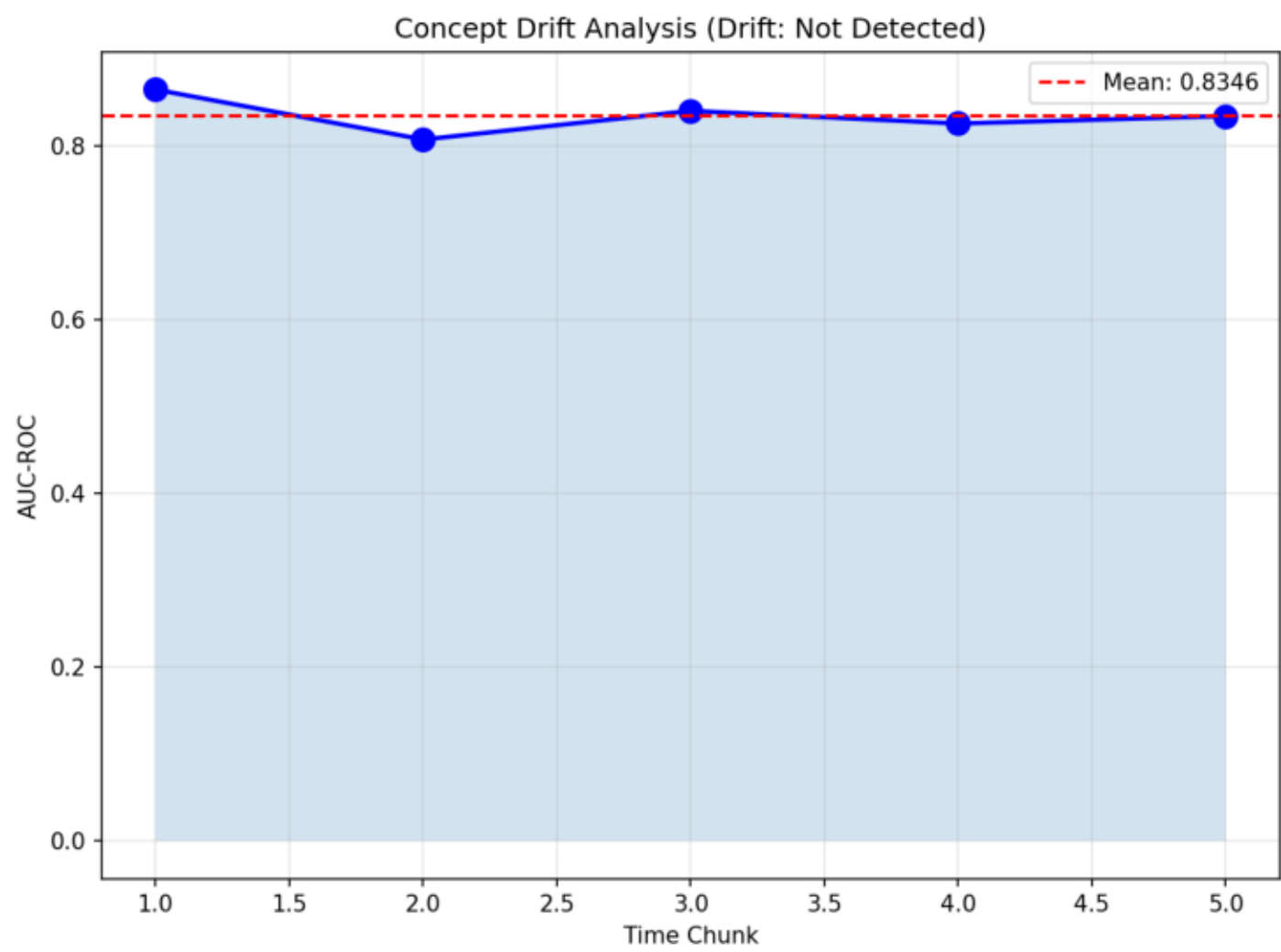
psi_distribution: Score distributions overlap consistent with PSI=0.0006.

3) Model Stability



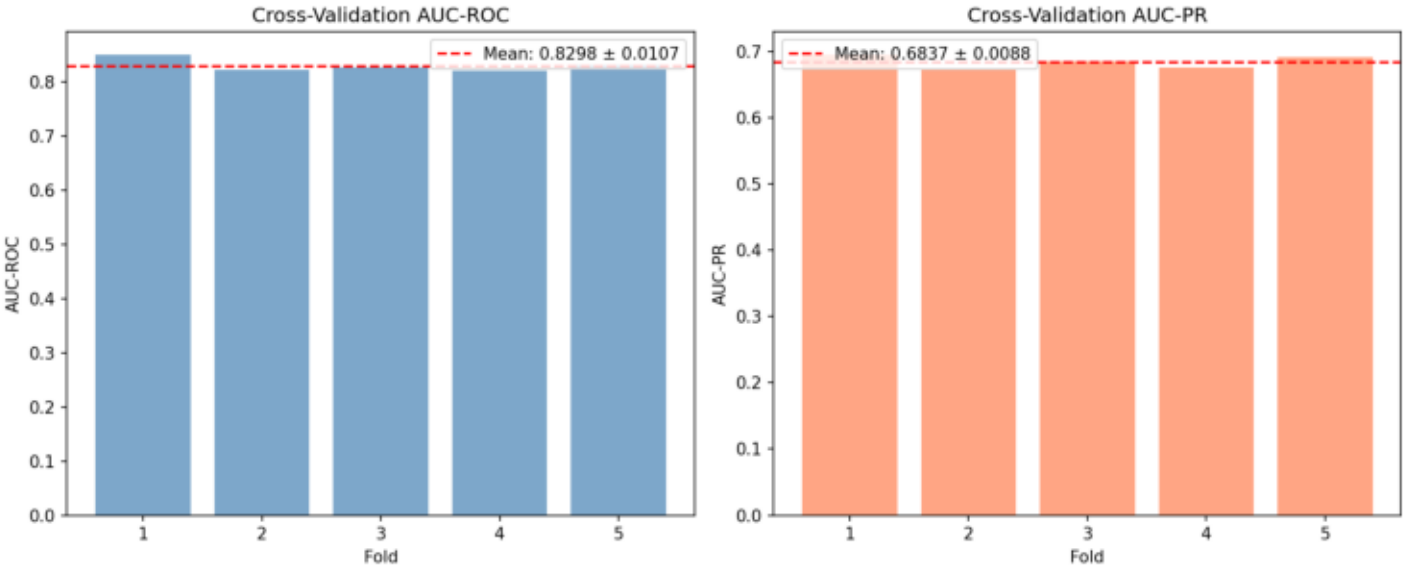
data_drift_heatmap: 0/18 features show drift; red bars highlight flagged features.

3) Model Stability



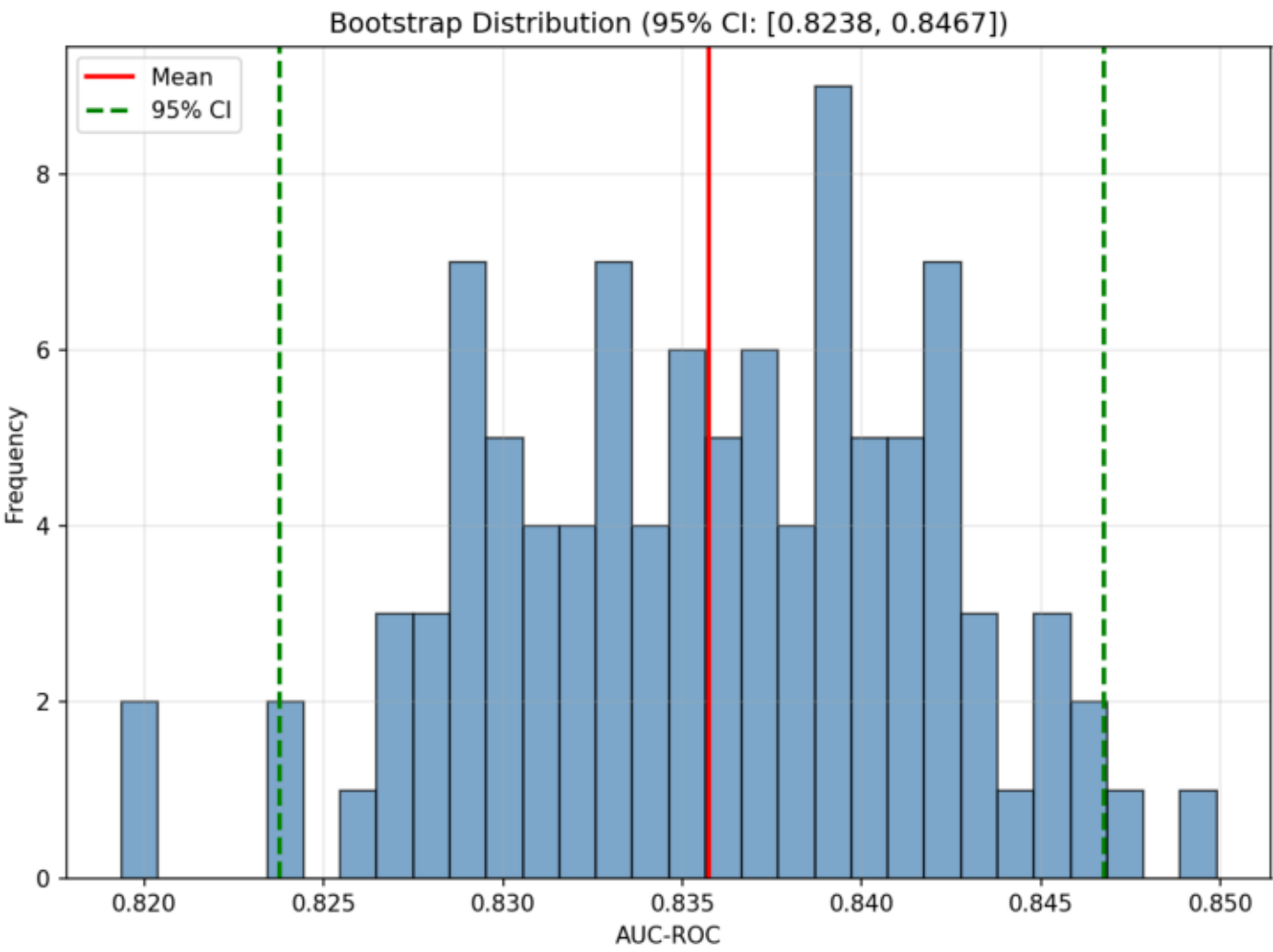
concept_drift: AUC across time chunks shows whether performance is stable over time.

3) Model Stability



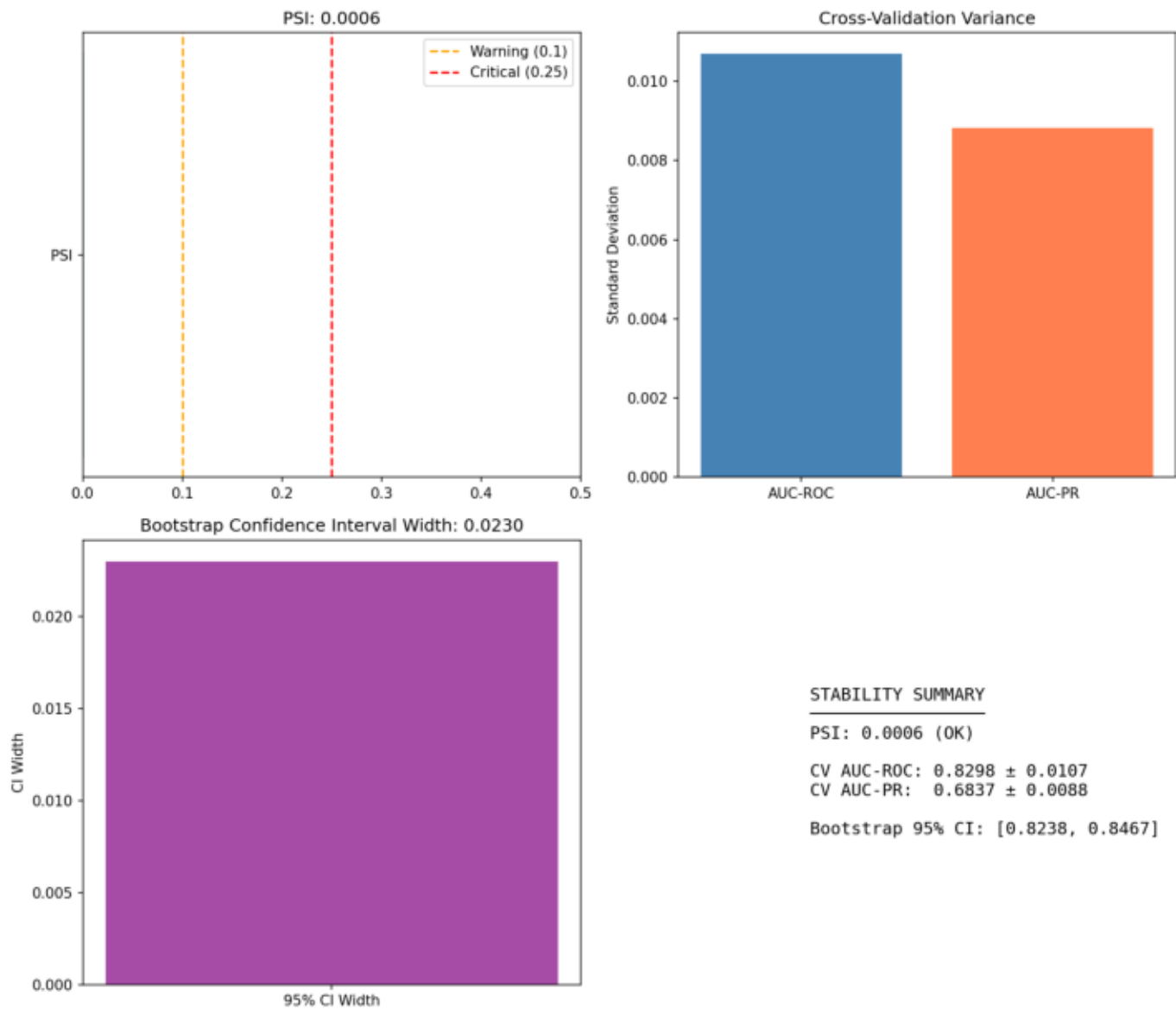
cv_results: Fold scores cluster around ROC 0.8298 and PR 0.6837.

3) Model Stability



bootstrap_distribution: Bootstrap CI [0.8238, 0.8467] reflects performance stability.

3) Model Stability



stability_summary: Dashboard summarizes PSI, CV variance, and bootstrap CI width.

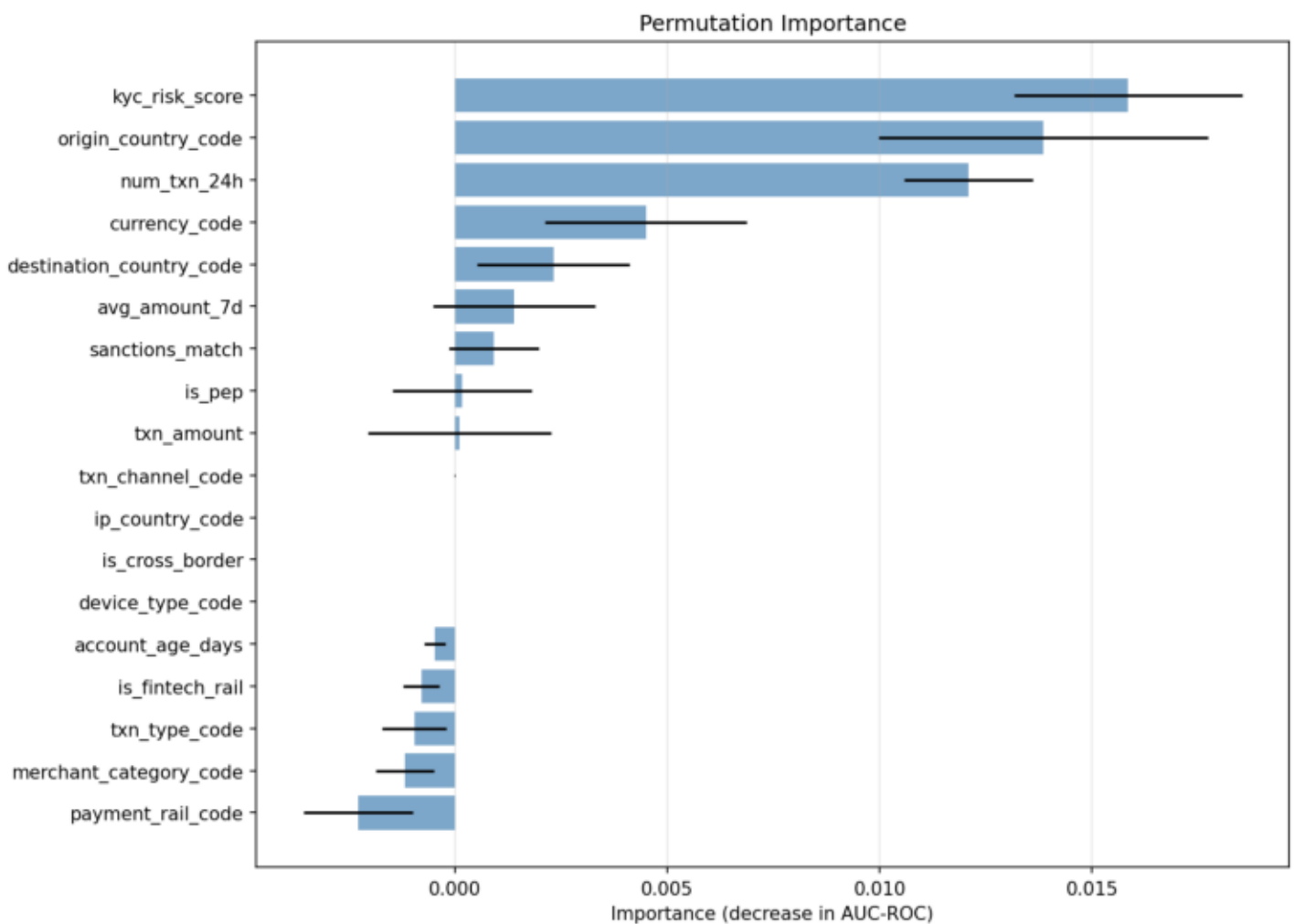
4) Model Interpretability (1/2)

Metric	Figure	Explanation
model_type	tree	Identifies the estimator family so explanation methods can be interpreted in the right context. Next: confirm the chosen explanation methods are appropriate for this model class.
methods_used	permutation, pdp, ice, shap, lime	Lists the interpretability methods produced in this run. Next: compare signals across methods before using them in documentation or model challenge.
perm_top_features	kyc_risk_score, or igin_country_code, num_txn_24h, currency_code, des tination_country_c ode ...	Shows features with the largest performance drop when permuted, which highlights global importance. Next: validate that the top drivers align with business logic, feature governance, and policy expectations.
shap_top_features	origin_country_cod e, num_txn_24h, kyc_risk_score, currency_code, txn_amount ...	Shows the most influential features under SHAP-based contribution analysis. Next: inspect whether directionality and ranking remain sensible across key segments.
lime_instances	3	Counts the locally explained observations reviewed with LIME. Next: examine representative true-positive and false-positive cases to understand local decision logic.

4) Model Interpretability (2/2)

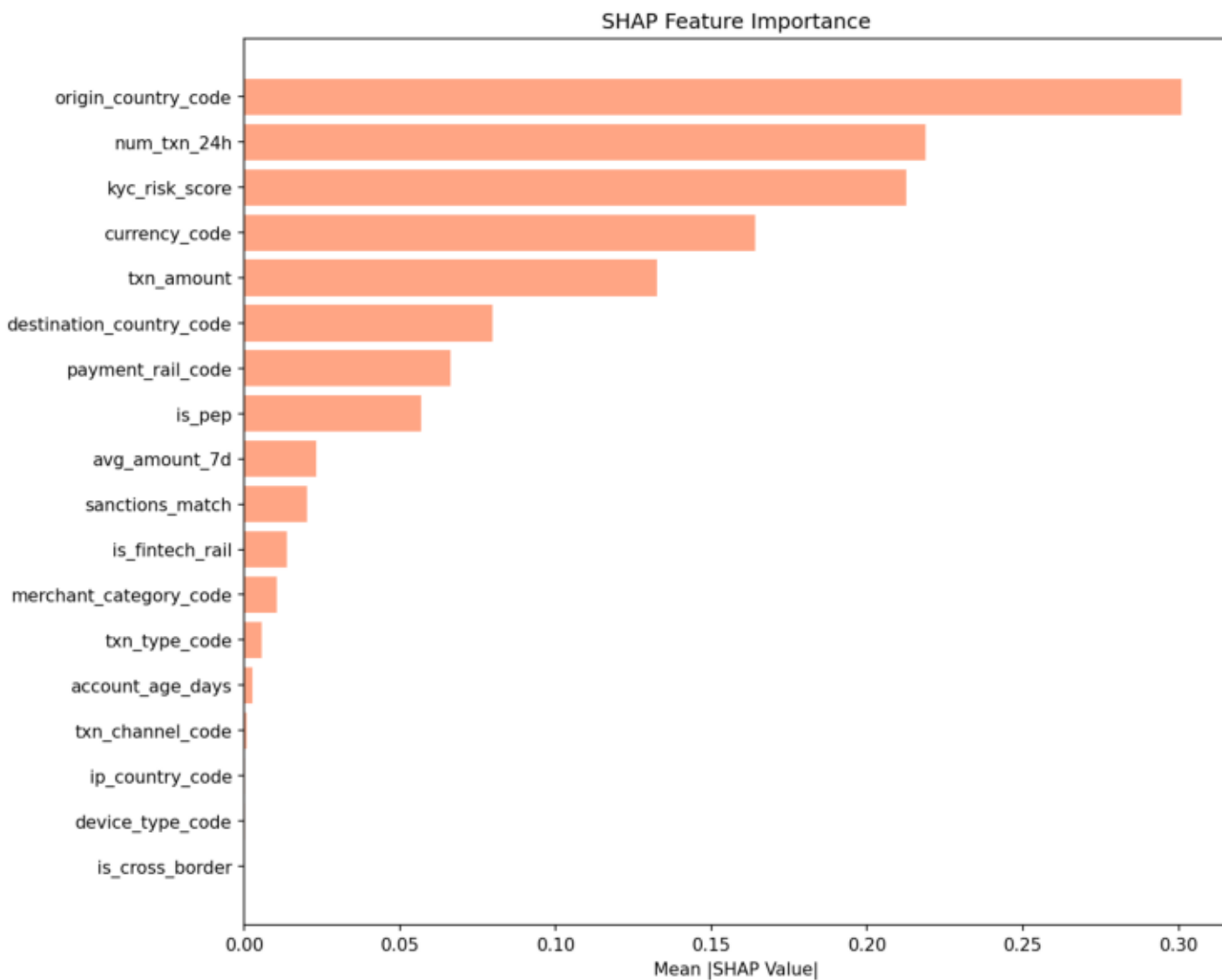
Metric	Figure	Explanation
pdp_features	txn_amount, account_age_days, avg_amount_7d, kyc_risk_score, num_txn_24h ...	Lists features with partial-dependence plots. Next: review whether average model response is monotonic, stable, and operationally sensible.
ice_features	txn_amount, account_age_days, avg_amount_7d, kyc_risk_score	Lists features with individual conditional expectation plots. Next: inspect heterogeneity and interactions that may be hidden in average-only views.

4) Model Interpretability



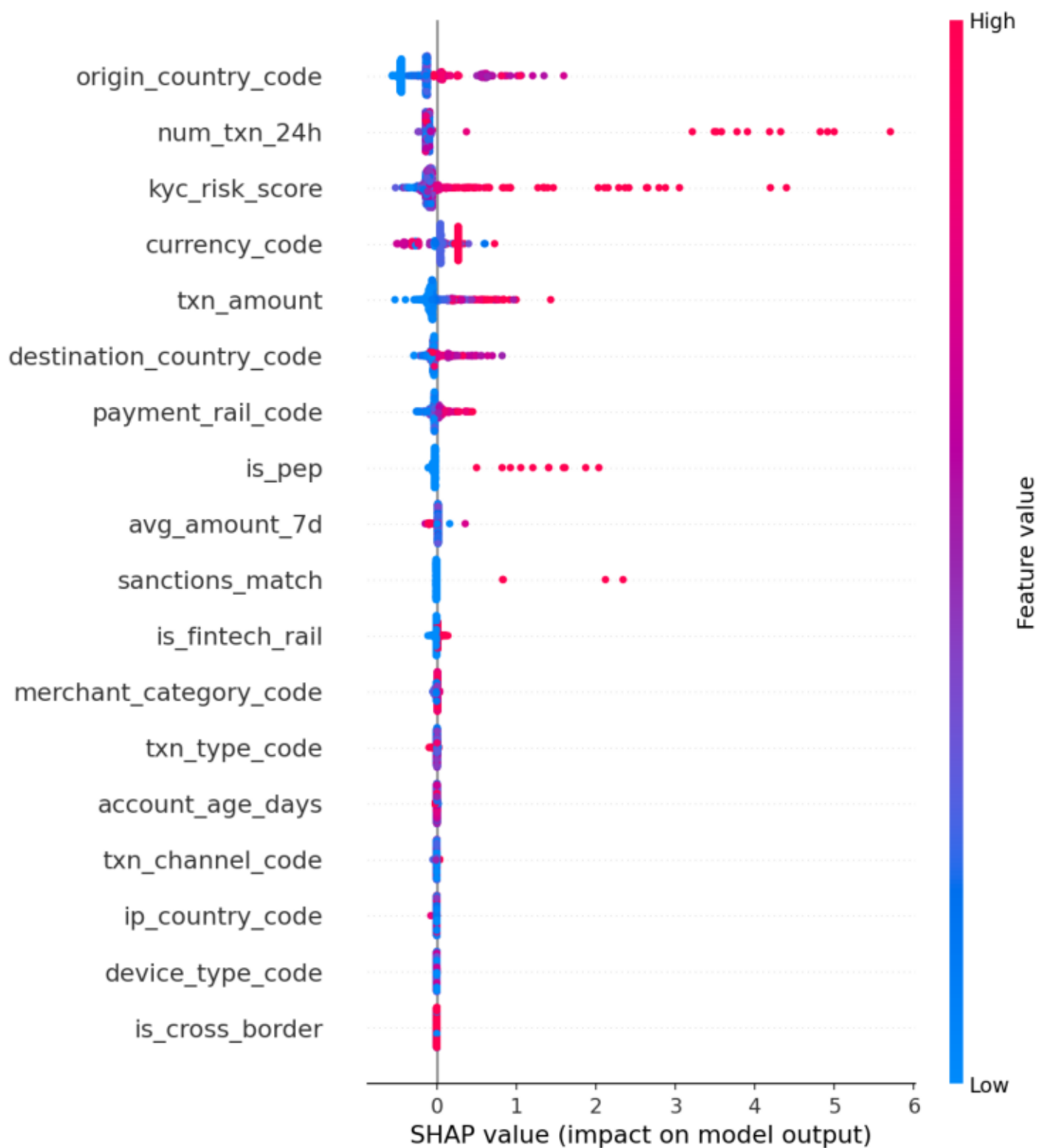
permutation_importance: Bars show how much AUC drops when each feature is permuted.

4) Model Interpretability

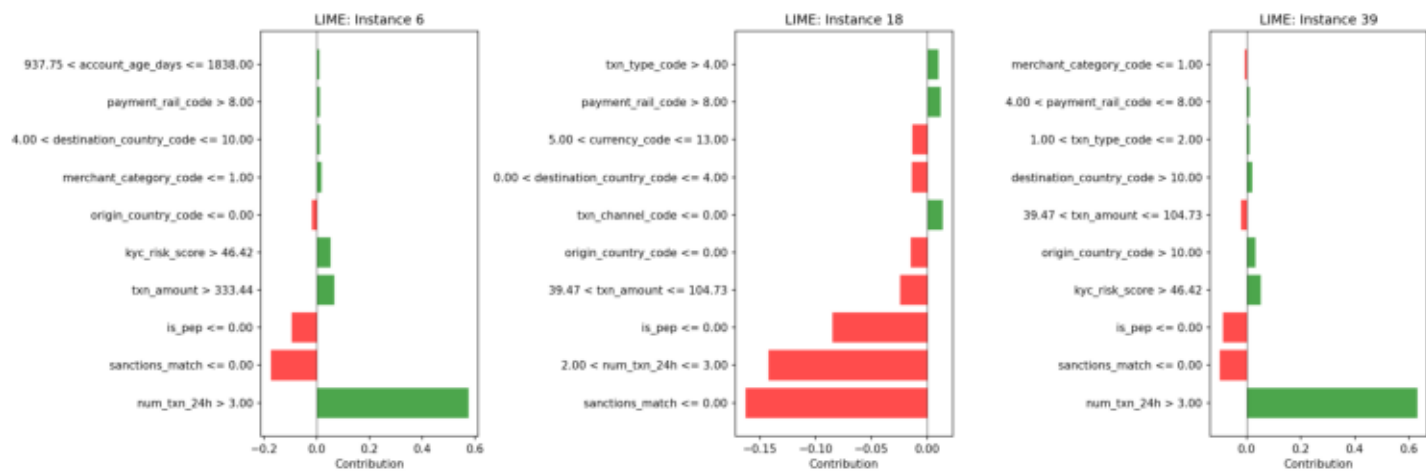


shap_bar: SHAP bar plot ranks global feature impact by mean absolute SHAP values.

4) Model Interpretability

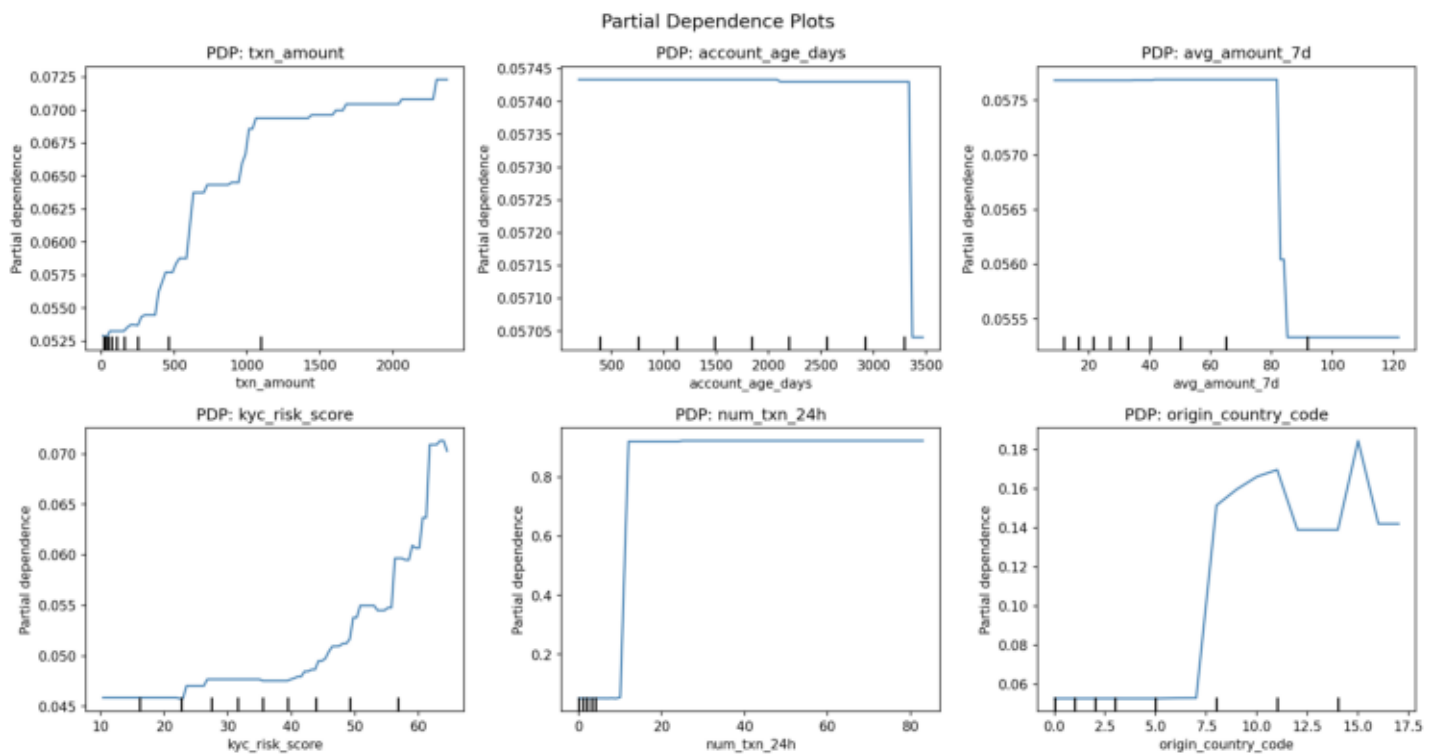


4) Model Interpretability



lime_explanation: LIME bars show local feature contributions for selected instances.

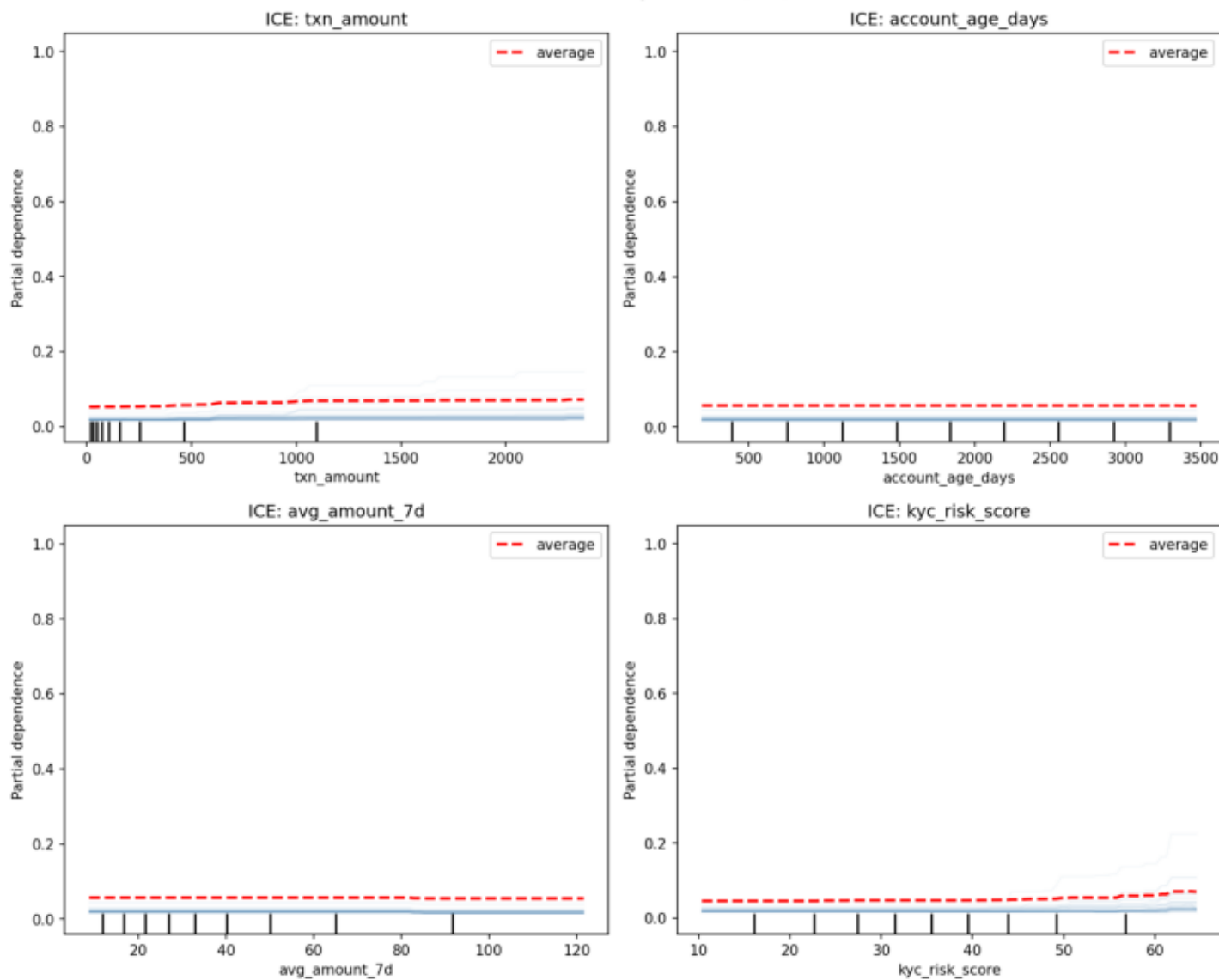
4) Model Interpretability



pdp: PDP curves show average model response as a feature varies.

4) Model Interpretability

Individual Conditional Expectation (ICE) Plots



ice: ICE curves show per-instance responses as a feature varies.